Variační principy

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Variační principy

Tvůj lagrangián, tvůj hood!

- svět se chová optimalizovaně
- požadujeme Lorentzovskou invarianci Lagrangeovy funkce

d'Alembertův princip, LRI

- v klasické mechanice $\left(f_i - \frac{dp_i}{dt}\right)\delta x^i = 0$, kde δx^i je izochronní ($\delta t \equiv 0$) virtuální posunutí
- pro systém s vazbou $\Phi(x^i, t) = 0$ s ní musí být virtuální posunutí ve shodě, tedy $\Phi_{,i}\,\delta x^i = 0$
  $\Rightarrow$ virtuální práce vazbových sil je nulová
- z d'Alembertova principu a vazby získáme LRI:

$$\frac{dp_i}{dt} = f_i + \lambda \Phi_{,i}$$

- v STR již nemohou být virtuální posunutí izochronní (nešlo by to zařídit vůči všem IS kvůli relativitě současnosti), od souřadnicového času t přecházíme k vlastnímu času τ, jinak je vše stejné:

$$\Phi_{,\mu}\,\delta x^\mu = 0 \qquad \left(F_\mu - \frac{dp_\mu}{d\tau}\right)\delta x^\mu = 0$$

$$\text{LRI:} \qquad \frac{dp_\mu}{d\tau} = F_\mu + \lambda \Phi_{,\mu}$$

$\Rightarrow$ 4 rovnice pro 6 neznámých: λ, x^μ, m_0

$+ \quad \Phi(x^\mu) = 0 \quad$ a $\quad \eta_{\mu\nu} u^\mu u^\nu = -c^2$

Hamiltonův princip, LRII

- klasický Hamiltonův princip:

systémy, kde $\exists V = V(q^i, \dot q^i, t)$ a holonomní vazby

$$S \equiv \int_{t_1}^{t_2} L(q^i, \dot q^i, t)\,dt$$

$$\delta S = 0 \quad \Rightarrow \quad \frac{d}{dt}\left(\frac{\partial L}{\partial \dot q^i}\right) - \frac{\partial L}{\partial q^i} = 0$$

STR jsou dva podstatné rozdíly:

- $\mathcal{L}$ musí být invariantem Lorentzovy transformace
  $\Rightarrow \mathcal{L} \neq T - V$, jelikož T je vůči různým IS různá, uvažujme ale stále $\mathcal{L} = \mathcal{L}(x^\mu, u^\mu)$
  $\rightarrow$ kvůli invarianci akce se musí integrovat přes $d\tau$
- při variaci přes světočáry jsou všechny virtuální světočáry parametrizovány různým vlastním časem
  $\Rightarrow$ při variaci (přechodu mezi světočárami) $x^\mu \rightarrow x^{*\mu} \equiv x^\mu + \delta x^\mu$ se musí variovat i vlastní čas, a tedy i jeho přírůstek $d\tau$:

$$d\tau^* = \frac{1}{c}\sqrt{-\eta_{\mu\nu}\,dx^{*\mu}\,dx^{*\nu}}$$

$$= \frac{1}{c}\sqrt{-\eta_{\mu\nu}(dx^\mu + d\delta x^\mu)(dx^\nu + d\delta x^\nu)}$$

$$= \frac{1}{c}\sqrt{c^2 d\tau^2 - 2\eta_{\mu\nu}\,dx^\mu d\delta x^\nu + O(\delta^2)}$$

$$= d\tau\left(1 - \frac{1}{c^2}\eta_{\mu\nu}\frac{dx^\mu}{d\tau}\frac{d\delta x^\nu}{d\tau} + O(\delta^2)\right)$$

$$= d\tau - \frac{1}{c^2}\eta_{\mu\nu}u^\mu d\delta x^\nu + O(\delta^2)$$

$$\Rightarrow \quad \delta d\tau \equiv d\tau^* - d\tau = -\frac{1}{c^2}\eta_{\mu\nu}u^\mu d\delta x^\nu$$

$$= -\frac{1}{c^2}u_\nu d\delta x^\nu$$

$$u^{*\mu} \equiv \frac{dx^{*\mu}}{d\tau^*} = \frac{dx^\mu + d\delta x^\mu}{d\tau - \frac{1}{c^2}u_\nu d\delta x^\nu} =$$

$$= \left(u^\mu + \frac{d\delta x^\mu}{d\tau}\right)\left(1 + \frac{1}{c^2}u_\nu\frac{d\delta x^\nu}{d\tau}\right) + O(\delta^2)$$

$$= u^\mu + \frac{d\delta x^\mu}{d\tau} + \frac{1}{c^2}u^\mu u_\nu\frac{d\delta x^\nu}{d\tau} + O(\delta^2)$$

$$= u^\mu + \left(\delta^\mu_\nu + \frac{1}{c^2}u^\mu u_\nu\right)\frac{d\delta x^\nu}{d\tau} + O(\delta^2)$$

$$\Rightarrow \quad \delta u^\mu = \left(\delta^\mu_\nu + \frac{1}{c^2}u^\mu u_\nu\right)\frac{d\delta x^\nu}{d\tau}$$

- platí $\delta u^\mu \perp u^\mu$, jelikož ( ) je projektor na třírozměrnou nadplochu kolmou k u^μ, ukažme:

$$\left(\delta^\mu_\alpha + \frac{1}{c^2}u^\mu u_\alpha\right)\left(\delta^\alpha_\nu + \frac{1}{c^2}u_\nu u^\alpha\right)$$

$$= \delta^\mu_\nu + \frac{1}{c^4}u^\mu(-c^2)u_\nu + \frac{1}{c^2}u^\mu u_\nu$$

$$+ \frac{1}{c^2}u^\mu u_\nu = \delta^\mu_\nu + \frac{1}{c^2}u^\mu u_\nu$$

$$\left(\delta^\mu_\nu + \frac{1}{c^2}u^\mu u_\nu\right)u^\nu = u^\mu - u^\mu = 0$$

$$\left(\delta^\mu_\nu + \frac{1}{c^2}u^\mu u_\nu\right)u_\mu = u_\nu - u_\nu = 0$$

$\Rightarrow$ je tenzor

závorku:

$$\eta_{\mu\nu}u^{*\mu}u^{*\nu} = (u_\nu + \delta u_\nu)(u^\nu + \delta u^\nu)$$

$$= -c^2 + 2u_\nu\delta u^\nu + O(\delta^2) = -c^2 + O(\delta^2)$$

$\Rightarrow u^{*\mu}$ je správně normalizovaný!

$$\Rightarrow \quad 0 = \delta S = \delta\int_{\tau_1}^{\tau_2}\mathcal{L}(x^\mu, u^\mu)\,d\tau = \int_{\tau_1}^{\tau_2}\delta\mathcal{L}\,d\tau + \mathcal{L}\,\delta d\tau$$

$$= \int_{\tau_1}^{\tau_2}\left[\frac{\partial\mathcal{L}}{\partial x^\mu}\delta x^\mu + \frac{\partial\mathcal{L}}{\partial u^\mu}\delta u^\mu\right]d\tau + \mathcal{L}\,\delta d\tau$$

$$= \int_{\tau_1}^{\tau_2}\left[\left(\frac{\partial\mathcal{L}}{\partial x^\mu}\delta x^\mu + \frac{\partial\mathcal{L}}{\partial u^\mu}\left(\delta^\mu_\nu + \frac{1}{c^2}u^\mu u_\nu\right)\frac{d\delta x^\nu}{d\tau}\right)d\tau - \mathcal{L}\frac{1}{c^2}u_\nu\,d\delta x^\nu\right]$$

$$= \int_{\tau_1}^{\tau_2}\left(\frac{\partial\mathcal{L}}{\partial x^\mu}\delta x^\mu + \left[\frac{\partial\mathcal{L}}{\partial u^\mu}\left(\delta^\mu_\nu + \frac{1}{c^2}u^\mu u_\nu\right) - \frac{\mathcal{L}}{c^2}u_\nu\right]\frac{d\delta x^\nu}{d\tau}\right)d\tau$$

$$\overset{PP}{=} \int_{\tau_1}^{\tau_2}\left(\frac{\partial\mathcal{L}}{\partial x^\mu}\delta x^\mu - \frac{d}{d\tau}\left[\frac{\partial\mathcal{L}}{\partial u^\mu}\left(\delta^\mu_\nu + \frac{1}{c^2}u^\mu u_\nu\right) - \frac{\mathcal{L}}{c^2}u_\nu\right]\delta x^\nu\right)d\tau$$

$$= \int_{\tau_1}^{\tau_2}\left(\frac{\partial\mathcal{L}}{\partial x^\nu} - \frac{d}{d\tau}\left[\frac{\partial\mathcal{L}}{\partial u^\mu}\left(\delta^\mu_\nu + \frac{1}{c^2}u^\mu u_\nu\right) - \frac{\mathcal{L}}{c^2}u_\nu\right]\right)\delta x^\nu d\tau$$

$$\Rightarrow \quad \frac{\partial\mathcal{L}}{\partial x^\nu} - \frac{d}{d\tau}\left[\frac{\partial\mathcal{L}}{\partial u^\nu} + \frac{u_\nu}{c^2}\left(\frac{\partial\mathcal{L}}{\partial u^\mu}u^\mu - \mathcal{L}\right)\right] = 0$$

pro holonomní vazbu je pak nutné uvažovat dále v rovnici $\delta S = 0$ člen:

$$\lambda\int_{\tau_1}^{\tau_2}\Phi_{,\nu}\,\delta x^\nu d\tau = 0$$

$\Rightarrow$ v LRII získáme člen $+\lambda\Phi_{,\nu}$

Nalezení Lagrangeovy funkce

- neexistuje žádný univerzální postup
- $\mathcal{L}$ je omezena požadavkem na Lorentzovskou invarianci
- někdy ho lze určit kombinací d'Alembertova a Hamiltonova variačního principu:

platí: $p_\mu d\delta x^\mu = m_0 u_\mu d\delta x^\mu = -m_0 c^2 d\delta\tau$

Integrací d'Alemberta pak:

$$0 = \int_{\tau_1}^{\tau_2}\left(F_\mu - \frac{dp_\mu}{d\tau}\right)\delta x^\mu d\tau$$

$$\overset{PP}{=} \int_{\tau_1}^{\tau_2}\left(F_\mu\delta x^\mu + p_\mu\frac{d\delta x^\mu}{d\tau}\right)d\tau$$

$$= \int_{\tau_1}^{\tau_2}\left(F_\mu\delta x^\mu d\tau - m_0c^2 d\delta\tau\right)$$

$\updownarrow$ porovnejme s

$$0 = \delta S = \int_{\tau_1}^{\tau_2}\left(\delta\mathcal{L}\,d\tau + \mathcal{L}\,d\delta\tau\right)$$

$\Rightarrow$ pro volnou částici: ($F^\mu = 0$) lze volit např. $\mathcal{L} = -m_0c^2$, jinak je snahou upravit do tvaru:

$$\delta\int_{\tau_1}^{\tau_2}(\text{NĚCO})\,d\tau \xrightarrow[\text{platit}]{\text{by mělo}} \text{NĚCO} \propto \mathcal{L}$$

Lagrangián nabité částice v EMP

$$F_{L\mu} = qF_{\mu\nu}u^\nu = q(A_{\nu,\mu} - A_{\mu,\nu})u^\nu$$

a platí: $F_{L\mu}\delta x^\mu = q(A_{\nu,\mu} - A_{\mu,\nu})u^\nu\delta x^\mu$

$$= q\left(\delta A_\nu u^\nu - \frac{dA_\mu}{d\tau}\delta x^\mu\right)$$

$$\Rightarrow \int_{\tau_1}^{\tau_2}\left[F_{L\mu}\delta x^\mu d\tau - m_0c^2\delta d\tau\right] =$$

$$= \int_{\tau_1}^{\tau_2}\left[q\left(\delta A_\nu u^\nu - \frac{dA_\mu}{d\tau}\delta x^\mu\right)d\tau - m_0c^2\delta d\tau\right]$$

$$\overset{PP}{=} \int_{\tau_1}^{\tau_2}\left[q\left(\delta A_\nu u^\nu + A_\mu\frac{d\delta x^\mu}{d\tau}\right)d\tau - m_0c^2\delta d\tau\right]$$

$$= \int_{\tau_1}^{\tau_2}\left[q\left(\delta A_\nu u^\nu d\tau + A_\mu d\delta x^\mu\right) - m_0c^2\delta d\tau\right]$$

$$= \int_{\tau_1}^{\tau_2}\left[q\left(\delta A_\nu u^\nu d\tau + A_\mu\delta(u^\nu d\tau)\right) - m_0c^2\delta d\tau\right]$$

$$= \int_{\tau_1}^{\tau_2}\left[\delta q(A_\nu u^\nu d\tau) - m_0c^2\delta d\tau\right]$$

$$= \delta\int_{\tau_1}^{\tau_2}\left[qA_\nu u^\nu - m_0c^2\right]d\tau$$

$$\Rightarrow \quad \mathcal{L} = qA_\nu u^\nu - m_0c^2$$

Zkouška

$$\frac{\partial\mathcal{L}}{\partial x^\nu} = qA_{\sigma,\nu}u^\sigma$$

$$\frac{\partial\mathcal{L}}{\partial u^\mu} = qA_\sigma\delta^\sigma_\mu = qA_\mu$$

$$\left(\frac{\partial\mathcal{L}}{\partial u^\mu}u^\mu - \mathcal{L}\right) = m_0c^2$$

$$\left[\frac{\partial\mathcal{L}}{\partial u^\nu} + \frac{u_\nu}{c^2}\left(\frac{\partial\mathcal{L}}{\partial u^\mu}u^\mu - \mathcal{L}\right)\right] = qA_\nu + m_0u_\nu$$

$$= qA_\nu + p_\nu$$

$$\Rightarrow \quad qA_{\sigma,\nu}u^\sigma - \frac{d}{d\tau}\left[qA_\nu + p_\nu\right] = 0$$

$$\Rightarrow \quad \frac{dp_\nu}{d\tau} = qA_{\sigma,\nu}u^\sigma - qA_{\nu,\sigma}u^\sigma$$

$$= q\left(A_{\sigma,\nu} - A_{\nu,\sigma}\right)u^\sigma = qF_{\nu\sigma}u^\sigma$$

:) ✓

Klasičtější formulace Hamiltonova principu

- "méně poctivě" $\Rightarrow d\tau$ a u^μ mají význam vlastního času a čtyř-rychlosti pouze na skutečné světočáře, tedy až po dosazení do E-L rovnic
- pokud tedy $d\tau^* = d\tau$, pak

$$u^{*\mu} = \frac{dx^{*\mu}}{d\tau^*} = \frac{dx^\mu + d\delta x^\mu}{d\tau} = u^\mu + \frac{d\delta x^\mu}{d\tau}$$

$$\Rightarrow \quad \delta u^\mu = \frac{d\delta x^\mu}{d\tau}$$

ale $\quad \eta_{\mu\nu}u^{*\mu}u^{*\nu} = -c^2 + 2\eta_{\mu\nu}u^\mu\delta u^\nu + O(\delta^2)$

$$= -c^2 + 2u_\nu\frac{d\delta x^\nu}{d\tau} + O(\delta^2)$$

- "hvězdný tvar" LRII:

$$\delta S = \delta\int_{\tau_1}^{\tau_2}\mathcal{L}\,d\tau = \int_{\tau_1}^{\tau_2}\left[\frac{\partial\mathcal{L}}{\partial x^\mu}\delta x^\mu + \frac{\partial\mathcal{L}}{\partial u^\mu}\delta u^\mu\right]d\tau$$

$$= \int_{\tau_1}^{\tau_2}\left[\frac{\partial\mathcal{L}}{\partial x^\mu}\delta x^\mu + \frac{\partial\mathcal{L}}{\partial u^\mu}\frac{d\delta x^\mu}{d\tau}\right]d\tau$$

$$\overset{PP}{=} \int_{\tau_1}^{\tau_2}\left[\frac{\partial\mathcal{L}}{\partial x^\mu}\delta x^\mu - \frac{d}{d\tau}\left(\frac{\partial\mathcal{L}}{\partial u^\mu}\right)\delta x^\mu\right]d\tau$$

$$= \int_{\tau_1}^{\tau_2}\left[\frac{\partial\mathcal{L}}{\partial x^\mu} - \frac{d}{d\tau}\left(\frac{\partial\mathcal{L}}{\partial u^\mu}\right)\right]\delta x^\mu d\tau$$

$$\Rightarrow \quad \frac{\partial\mathcal{L}}{\partial x^\mu} - \frac{d}{d\tau}\left(\frac{\partial\mathcal{L}}{\partial u^\mu}\right) = 0$$

Odpovídající Lagrangián

- jiná formulace δ úlohy $\Rightarrow$ jiný lagrangián
- dobré tipy jsou:

$$\mathcal{L} = \frac{1}{2}m_0u_\sigma u^\sigma + qA_\sigma u^\sigma$$

$$\mathcal{L} = -m_0c\sqrt{-u_\sigma u^\sigma} + qA_\sigma u^\sigma$$

! Pozor $u_\sigma u^\sigma$ nelze zesktučnit na $-c^2$

$\Rightarrow$ lze až po zderivování v LRII

$$\frac{\partial\mathcal{L}}{\partial x^\mu} = qA_{\sigma,\mu}u^\sigma$$

$$\frac{\partial\mathcal{L}}{\partial u^\mu} = \begin{cases} m_0u_\mu + qA_\mu = p_\mu + qA_\mu \\ m_0c\dfrac{u_\mu}{\sqrt{-u_\sigma u^\sigma}} + qA_\mu = m_0u_\mu + qA_\mu \\ \qquad = p_\mu + qA_\mu \end{cases}$$

$$\Rightarrow \quad \frac{dp_\mu}{d\tau} = q\left(A_{\sigma,\mu}u^\sigma - A_{\mu,\sigma}u^\sigma\right) = qF_{\mu\sigma}u^\sigma$$

Variační odvození Maxwellových rovnic

- variační princip je správný i v teorii pole
  $\Rightarrow$ Euler měl pravdu – ve světě se neděje nic, na čem by se neprojevovala nějaká vlastnost minima či maxima
- z Hamiltonova variačního principu lze odvodit netriviální sérii Maxwellových
  $\rightarrow$ konfigurace je spojitá, uvažujme (vlastní) hustotu Lagrangeánu
  - rozměr hustoty energie
  - invariant
  - sestavena z EM veličin $\Rightarrow J^\mu, A_\mu, F_{\mu\nu}$
    $\Rightarrow$ invarianty $F_{\alpha\beta}F^{\alpha\beta}$, $J^\alpha A_\alpha$
- správně vsadíme:

$$\mathcal{L} = \int_{V_0}\left[-\frac{1}{4\mu}F^{\alpha\beta}F_{\alpha\beta} + J^\alpha A_\alpha\right]dV_0$$

$\Rightarrow$ hledáme konfiguraci pole v pevně zadané prostoročasové oblasti Ω s OP na $\partial\Omega \rightarrow \delta A_\alpha|_{\partial\Omega} = 0$, zdroje J^α jsou pevně zadané!

$\rightarrow$ variujeme se pouze vůči δA_α:

$$\delta S = \delta\int_{\tau_1}^{\tau_2}\int_{V_0}\left[-\frac{1}{4\mu}F^{\alpha\beta}F_{\alpha\beta} + J^\alpha A_\alpha\right]dV_0\,d\tau$$

$$= \int_\Omega\left[-\frac{1}{2\mu}F^{\alpha\beta}\delta F_{\alpha\beta} + J^\alpha\delta A_\alpha\right]d\Omega$$

$$= \int_\Omega\left[-\frac{1}{2\mu}F^{\alpha\beta}\left(\delta A_{\beta,\alpha} - \delta A_{\alpha,\beta}\right) + J^\alpha\delta A_\alpha\right]d\Omega$$

$$= \int_\Omega\left[\frac{1}{\mu}F^{\alpha\beta}\delta A_{\alpha,\beta} + J^\alpha\delta A_\alpha\right]d\Omega$$

$$\overset{PP}{=} \int_\Omega\left[\frac{-1}{\mu}F^{\alpha\beta}{}_{,\beta}\,\delta A_\alpha + J^\alpha\delta A_\alpha\right]d\Omega$$

$$= \int_\Omega\left[-\frac{1}{\mu}F^{\alpha\beta}{}_{,\beta} + J^\alpha\right]\delta A_\alpha\,d\Omega$$

$$\Rightarrow \quad F^{\alpha\beta}{}_{,\beta} = \mu J^\alpha$$

$$\delta\left(F^{\alpha\beta}F_{\alpha\beta}\right) = \delta F^{\alpha\beta}F_{\alpha\beta} + F^{\alpha\beta}\delta F_{\alpha\beta}$$

$$= \delta F_{\alpha\beta}F^{\alpha\beta} + F^{\alpha\beta}\delta F_{\alpha\beta} = 2F^{\alpha\beta}\delta F_{\alpha\beta}$$

$$F^{\alpha\beta}\left(\delta A_{\beta,\alpha} - \delta A_{\alpha,\beta}\right) = F^{\alpha\beta}\delta A_{\beta,\alpha}$$

$$- F^{\alpha\beta}\delta A_{\alpha,\beta} = F^{\beta\alpha}\delta A_{\alpha,\beta} - F^{\alpha\beta}\delta A_{\alpha,\beta}$$

$$= -2F^{\alpha\beta}\delta A_{\alpha,\beta}$$

$$\int_\Omega F^{\alpha\beta}\delta A_{\alpha,\beta}\,d\Omega = \int_\Omega\left(F^{\alpha\beta}\delta A_\alpha\right)_{,\beta}d\Omega$$

$$- \int_\Omega F^{\alpha\beta}{}_{,\beta}\,\delta A_\alpha\,d\Omega$$

$$\int_\Omega\left(F^{\alpha\beta}\delta A_\alpha\right)_{,\beta} \overset{\text{Gauss}}{=} \oint_{\partial\Omega}F^{\alpha\beta}\underset{0}{\underset{\parallel}{\delta A_\alpha}}\,d\Sigma_\beta = 0$$

- druhá sada Maxwellových plyne triviálně